

Rensselaer Polytechnic Institute

MANE 6140

Introduction to Computational Fluid Dynamics

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Analysis of Cable-in-Conduit

Superconducting Magnets

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1: Problem Description and Model (100% Pienaar)

Superconducting magnets are a critical feature of many advanced systems throughout the science and engineering community. The unique ability to create powerful force fields in space at distance is key in applications such as MRIs, magnetic levitation vehicles, particle accelerators, and fusion reactors. Inducing superconducting properties in a metal, however, is not a trivial task, and often involves holding a metal at temperatures close to absolute zero. In the context of fusion reactors, this is especially challenging, as magnets which work to control plasma moving at hundreds of million Kelvin must be held at 4-6 Kelvin. Currently, thermal management of these superconducting magnets is done by passing super cooled Helium (SHe) over the magnetic strands, in a configuration known as the Cable-in-Conduit.



Figure 1: Cable-in-Conduit [1]

Cable-in-Conduit (CIC) systems are made up of a central, spiralized, channel through which SHe flows. The spirals in the channel allow SHe to flow between the metal superconducting strands and the central channel. These strands are encased in a steel-jacket, and wrapped around a toroidal coil several hundred times to create the magnetic field necessary to contain the fusing plasma.

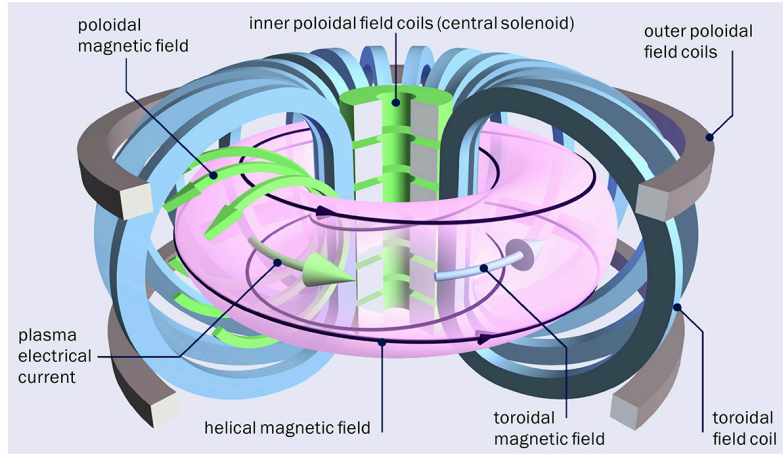


Figure 2: Tokamak Reactor [2]

This report will focus on a thermal analysis of the superconducting magnets in a representative CIC section, and seek to understand how the CIC system must be configured in order to maintain operating temperatures.

A numerical approach was chosen to conduct the desired thermal analysis. Siemens NX was used to create a representative 3D model of a CIC section consisting of a bounding box, an outer fluid body, magnetic strands, a central solid body, and a central fluid body.

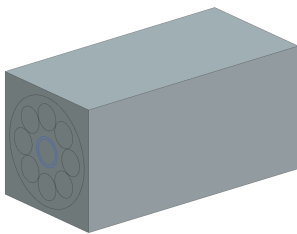


Figure 3: Bounding Box

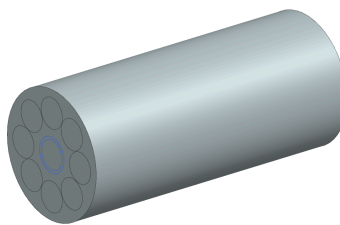


Figure 4: Outer Fluid Body

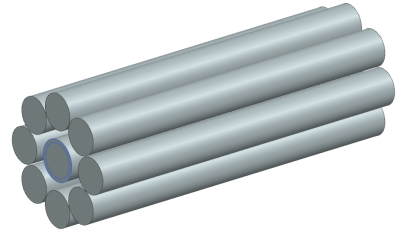


Figure 5: Magnetic Strands

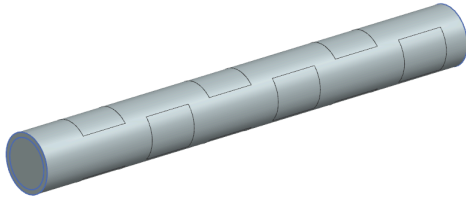


Figure 6: Central Solid Body

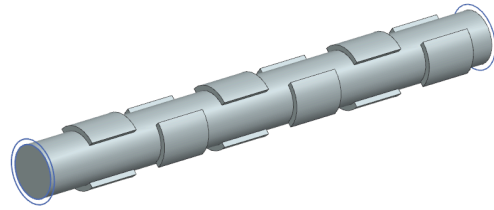


Figure 7: Central Fluid Body

This design improves over the midterm geometry, where only a central fluid body, and an outer magnetic body was considered, shown in Figure 8. Dimensions of our model are given in detail in Appendix A.

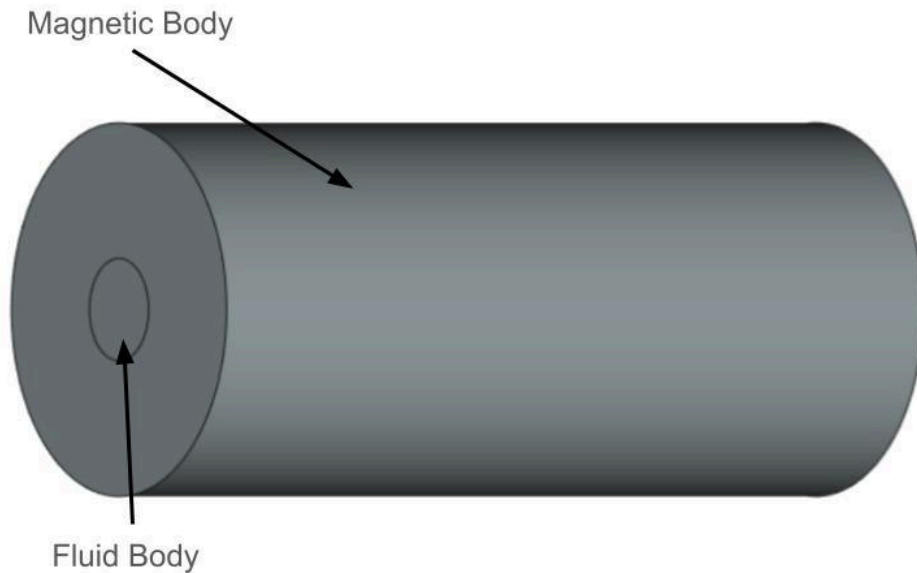


Figure 8: Midterm Geometry

A central goal of the geometry was to emulate the effects of the spiralized central channel of real world CIC systems, where coolant flowing through the central channel leaks out over the magnetic strands in the outer channel. This mixing of fluid from the inner and outer channel

significantly improves the cooling of the magnetic strands, helping keep them within operating temperatures.

In order to model the real world spiralized central channel, windows were “cut” out of our solid central channel. The central fluid body was then extruded to fill the gaps in the outer channel, shown in Figures 6 and 7, allowing for flow between the inner and outer fluid bodies.

The magnetic strands were modeled by creating eight, 10.5 mm diameter cylinders and placing them symmetrically around the central channel, shown in Figure 5. This leads to a porosity of 0.5, slightly higher than that taken by other similar projects of 0.37 [8][9]. Despite the high porosity, modeling the strands in this way was a straightforward way to induce an inner and outer fluid body, separate from the magnetic strands.

The magnetic strands, made up of two parts Niobium Tin and one part Copper, were given a weighted average of their thermal properties at 4 Kelvin. The coolant properties were taken as those of super cooled fluid helium at 4 Kelvin. The outer bounding box and inner central channel were taken to be stainless steel (316L). All physical properties are summarized in Table 1.

	Density ($\frac{kg}{m^3}$)	Thermal Conductivity ($\frac{w}{mK}$)	Heat Capacity $\frac{J}{kgK}$	Prandtl Number
SS316L	8000	0.15	0.5	NA
SHe	140.6	0.008852	5811	0.7878
Nb3Sn	8040	0.05615	1.206	NA
Cu	8960	3537	0.1250	NA
NB3Sn - Cu average	8347	1179	0.1005	NA

Table 1: Physical Properties [4] - [6]

2: Problem Set Up (100% Pienaar)

To accurately simulate the thermal-hydraulic behavior of the fluid bodies and internal strands, the following physical models and boundary conditions were implemented.

Flow was assumed incompressible and turbulent. Flow in the central fluid body was assumed to sinusoidally oscillate between $0.1 \frac{m}{s}$ and $1.0 \frac{m}{s}$ with a frequency of 4 Hz. This reflects the fact that coolant flow within the central channel is variable in real world systems, and can feasibly be within 0.1 and $1.0 \frac{m}{s}$ [10]. Flow in the outer fluid body was taken to be a constant value of $0.1 \frac{m}{s}$ to represent the forward momentum imparted on the fluid around the strands via the spiral perforation of real CICs. All fluid enters at a constant temperature of 4K.

The magnetic strands were each given a constant volumetric heating of $5000 \frac{W}{m^3}$. A no slip boundary condition was assumed for the circular ends of the magnetic strands and central solid, the central solid inner and outer surfaces which interface with fluid bodies, and the inner cylindrical face of the bounding box. Given that these magnetic coils exist within a vacuum chamber, all vacuum facing surfaces were assumed adiabatic.

The current model represents a substantial improvement over the midterm assumptions by introducing turbulent flow and the presence of multiple independent heating elements.

3: Numerical Scheme (50% Pienaar, 50% Rolleri)

Altair CFD 2024.1 was chosen to conduct the conjugate thermal-fluid modeling. The acusolve solver was employed with heat transfer calculations enabled, and both a transient and steady state run. The transient simulation is detailed below.

For the transient UDF simulation, the initial velocity V_0 was taken to be $0.5 \frac{m}{s}$. This value was selected as it aligns with the standard helium flow rates observed in tokamaks, which can range from $0.1 \frac{m}{s}$ to $1 \frac{m}{s}$ [10]. The radius of the central channel is 5 mm or 0.005 m. For the UDF, a sinusoidal frequency of 4 Hz was chosen so that the effects of the UDF could be easily identifiable in the results. The following steps were then taken to calculate the simulation's timestep, total duration, and total number of steps:

$$\text{Oscillation Period:} \quad 4 \text{ Hz} \rightarrow T_{\text{osc}} = 0.25 \text{ sec.} \quad (1)$$

$$\text{Omega:} \quad T_{\text{osc}} = \frac{2\pi}{\omega} \rightarrow \omega = 25.132741 \quad (2)$$

$$\text{Womersley Number:} \quad \omega = \frac{\gamma^2 \nu}{R^2} \rightarrow \gamma = \sqrt{\frac{\omega R^2}{\nu}} \rightarrow \gamma = 157.6 \quad (3)$$

$$\text{Time Step (dt):} \quad dt = \frac{T_{\text{osc}}}{16} = 0.0156231 \quad (4)$$

$$\text{Total Sim. Time:} \quad T_{\text{sim}} = 5(T_{\text{osc}}) = 1.25 \text{ sec.} \quad (5)$$

Number of Time Steps:
$$N_{\text{steps}} = \frac{T_{\text{sim}}}{dt} = 80 \text{ steps} \quad (6)$$

Following the configuration of simulation details, a mesh could be made. The mesh was constructed by applying both a surface mesh to most surfaces in the domain, as well as boundary layer refinement in select locations.

Surface meshing was applied to the exterior faces of the bounding box, the circular faces of solid and fluid bodies, and the cylindrical length of each solid and fluid cylindrical body. The surface mesh element size was taken to be the diameter describing a particular body divided by 16, and is summarized in Table 2.

Body	Characteristic Diameter	Surface Mesh Element Size
Central Fluid	10 mm	0.625
Central Solid	10 mm	0.625
Outer Fluid	43.7 mm	2.73125
Magnetic Strand	10.5 mm	0.65625
Bounding Box	49 mm	3.0625

Table 2: Surface Mesh Details

A boundary layer mesh was applied to the outer surface of the inner fluid body, the inner surface of the outer fluid body, which interfaces with the central solid, and at the solid-fluid interface of the magnetic strands and outer fluid body. The boundary mesh produced eight layers with constant growth, the ability to alter the surface mesh, and a mesh element size proportional to each body's diameter divided by 256, summarized in Table 3.

It should be noted that a boundary layer was applied both to the fluid interfacing with the magnetic strands, as well as the magnetic strands themselves, as it was important to understand if flow in the outer fluid body cools the strands despite their volumetric heating.

Body-Body interface	Characteristic Diameter	Surface Mesh Element Size
Central Fluid-Central Solid	10 mm	0.0390625
Central Solid-Outer Fluid	10 mm	0.0390625
Magnetic Strand-Outer Fluid	43.7 mm	0.041015625

Table 3: Boundary Layer Mesh Details

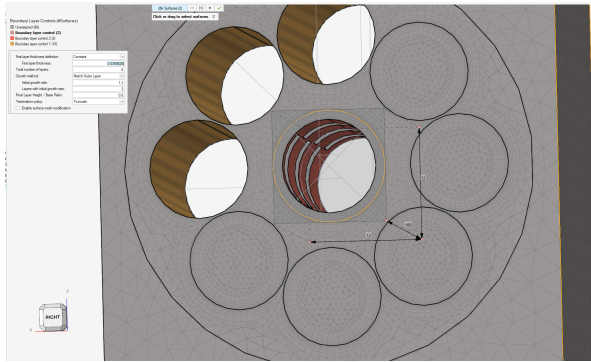


Figure 9: Boundary Layer on Outer Fluid Body and Central Solid Body

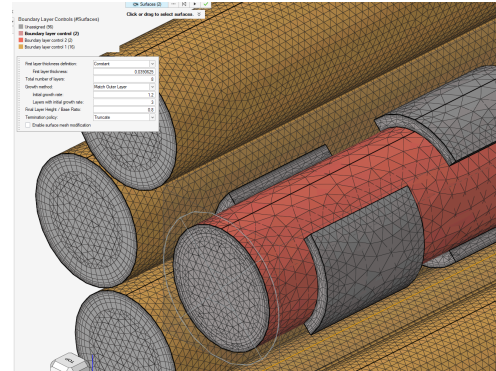


Figure 10: Boundary Layer on Magnetic Strands and Central Fluid Body

Finally a volumetric mesh was taken with an average element size of 2 mm, producing the final mesh.

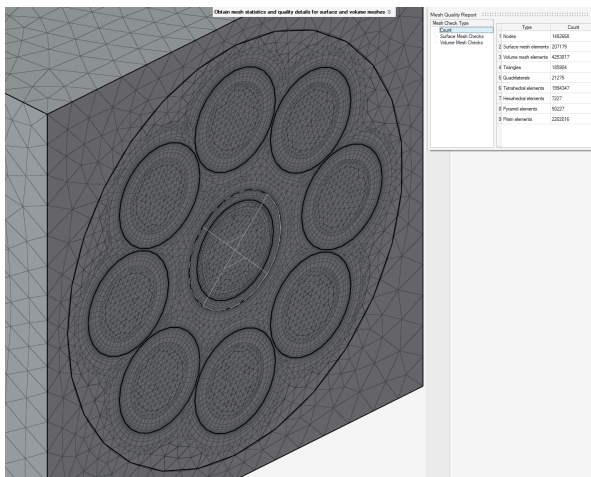


Figure 11: Final Mesh and Quality Report

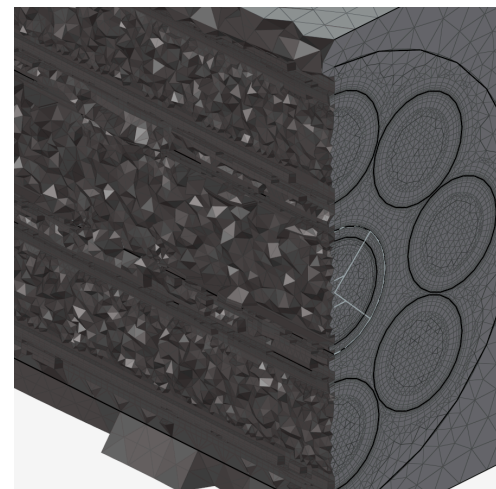


Figure 12: Cross Section of Mesh

4: Results (100% Roller)

Baseline Steady State Simulation:

The results of the steady state baseline analysis are shown and discussed below in Figures 13 through 16.

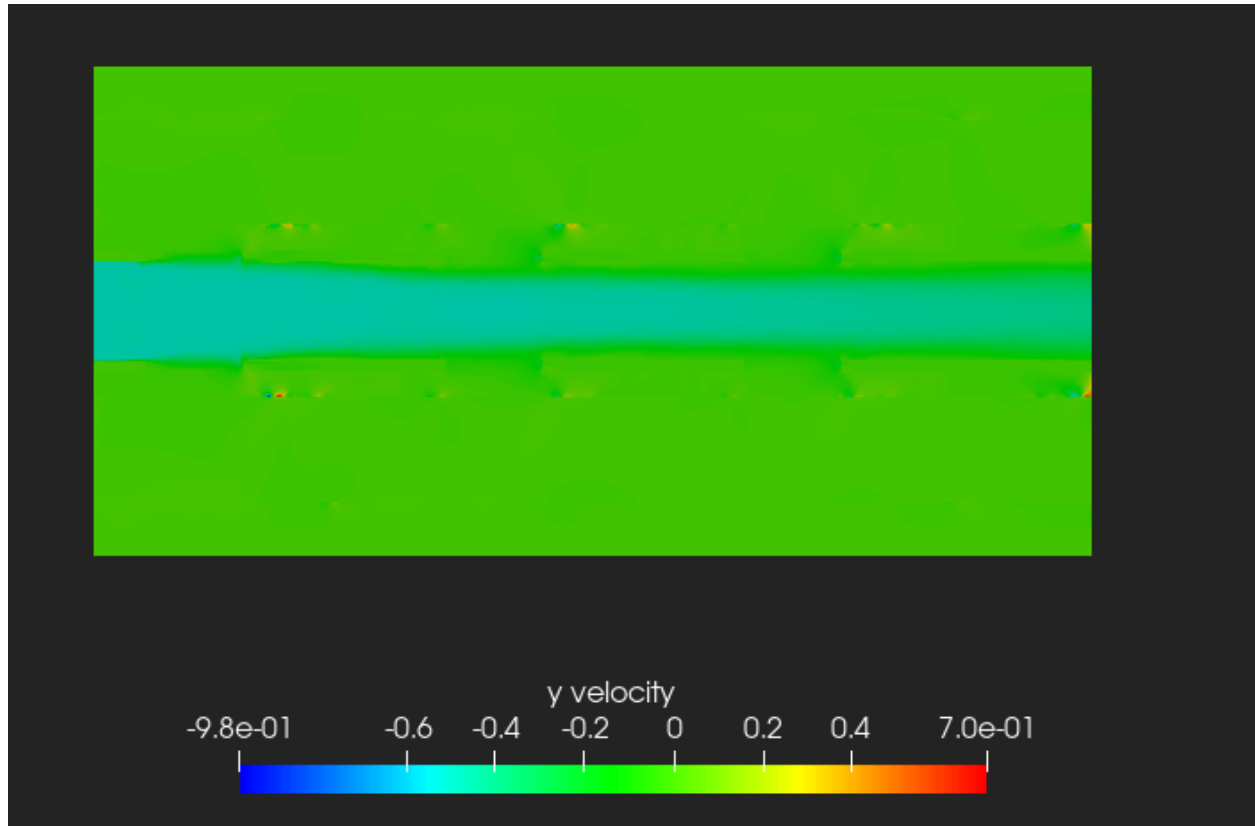


Figure 13: Steady State Y-Velocity Plot

The y velocity plot shows a strong inlet velocity of $0.5 \frac{m}{s}$ that decreases in velocity as the flow moves toward the outlet. There are two main regions where the velocity showed extremes. The first of these regions is near the outlets, on the right hand side of Figure 13. These minor spikes were attributed to backflow from the solver. However, the middle region of the plot shows red and yellow dots which were attributed to fluid acceleration between the strands. Given

the tight packing of these 8 strands within the model, the velocity was forced through the gaps between the strands causing localized spikes in the velocity.

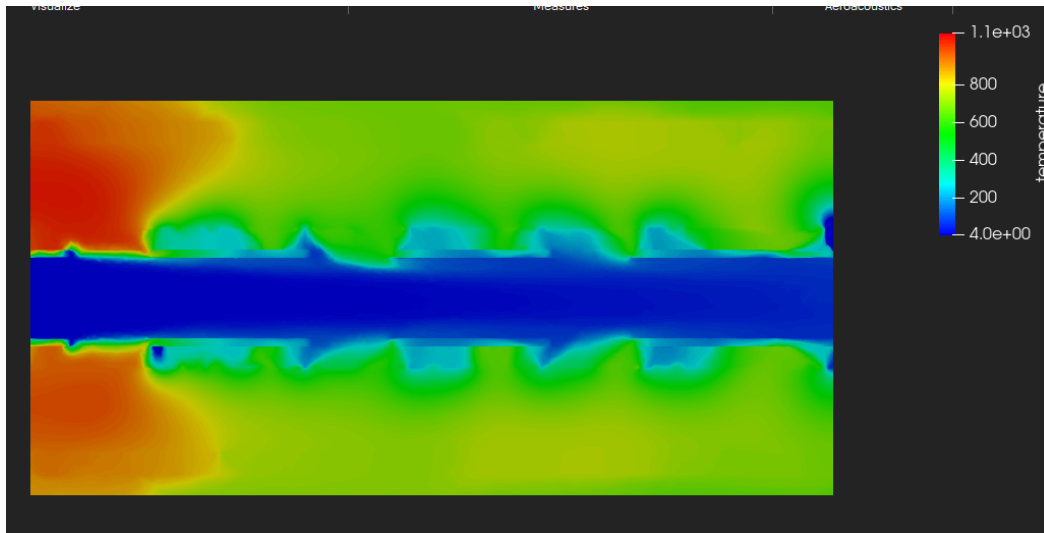


Figure 14: Steady State Temperature Plot

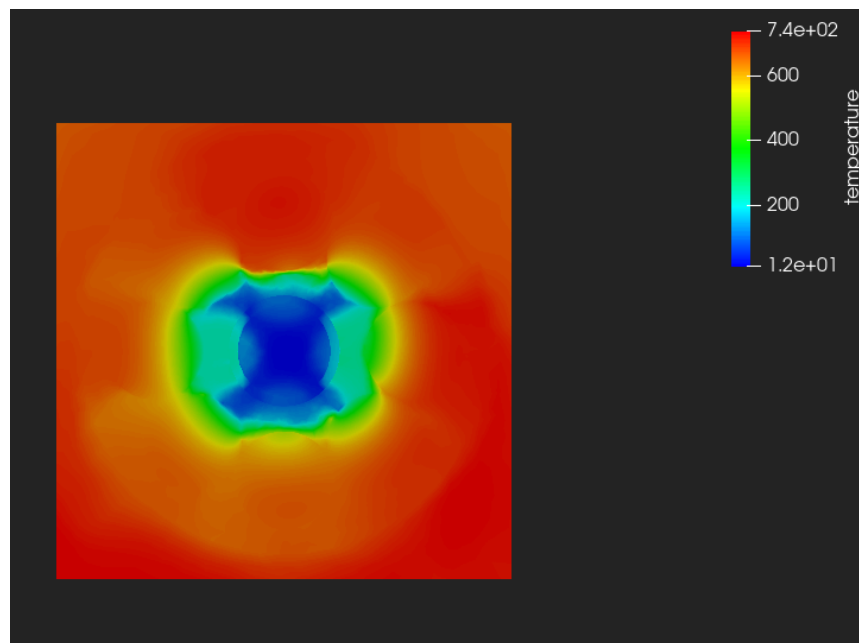


Figure 15: Cross Sectional Steady State Temperature Plot

The temperature results, shown in Figures 14 and 15, depict the evolution of the temperature within the CIC design. Figure 14 shows jetting of the incoming liquid helium coolant into the strand region of the model. Thermal energy is clearly being removed from the strands. The large high temperature region on the left of the plot was expected as a no slip BC was applied to the inlet side face of the fluid region surrounding the strands.

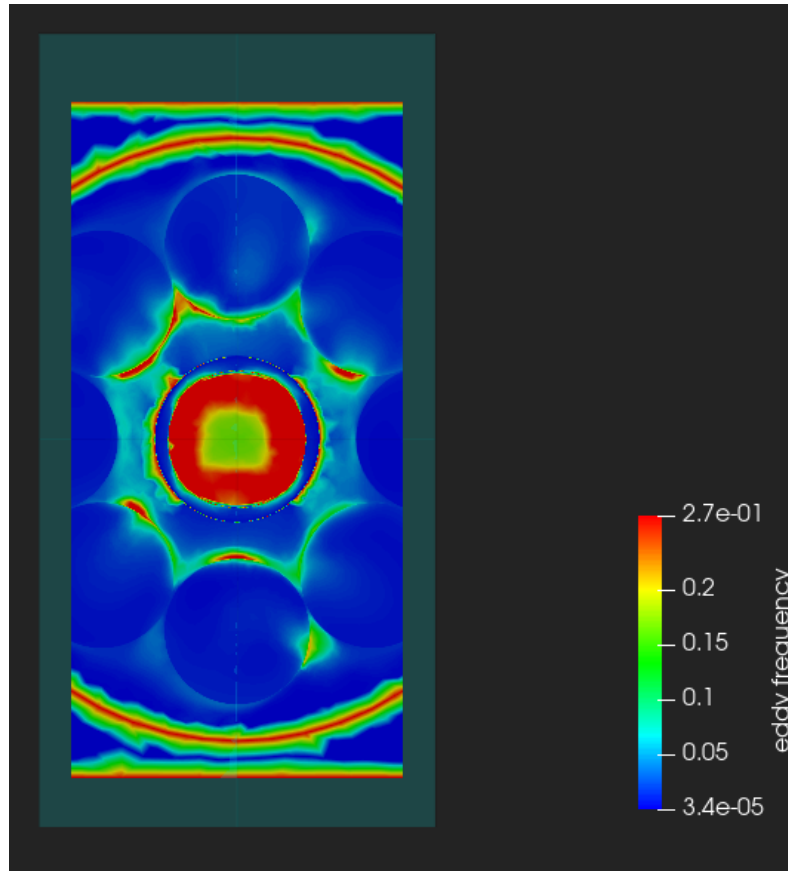


Figure 16: Cross Sectional Steady State Eddy Frequency Plot

Eddy frequency can be used to indirectly visualize the effectiveness of heat removal from bodies in the design. For example, the high eddy frequency near the strands indicates that heat removal will be highly efficient in those areas. This is due to very high local turbulence which will result in a sharp thermal gradient between the strands and the surrounding fluid, thereby increasing the heat removal rate.

Transient UDF Simulation:

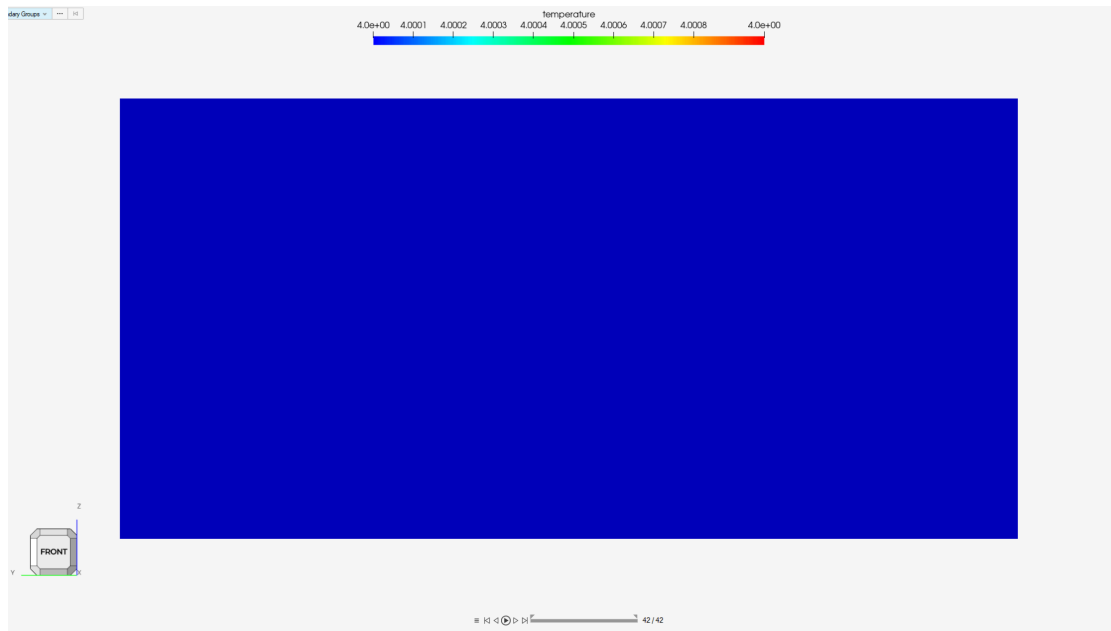


Figure 17: Lengthwise Slice of Temperature Results



Figure 18: Cross Sectional Slice of Temperature Results

The thermal results can be seen in Figures 17 and 18. It is immediately obvious that the thermal gradient is very weak. This behavior was expected and is explained in the analysis section.

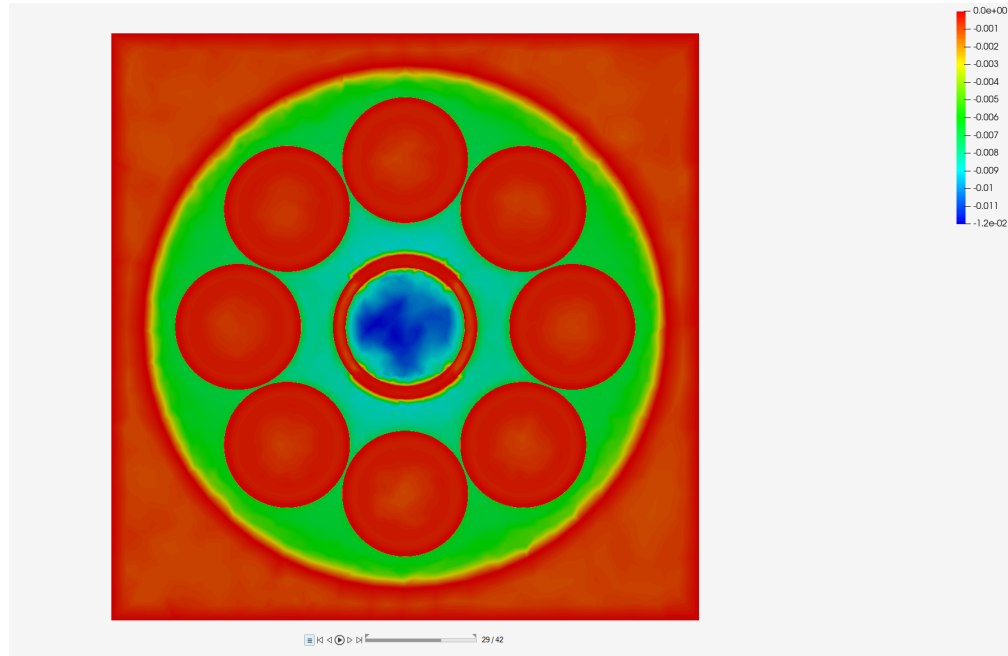


Figure 19: Cross Sectional Slice of Y velocity

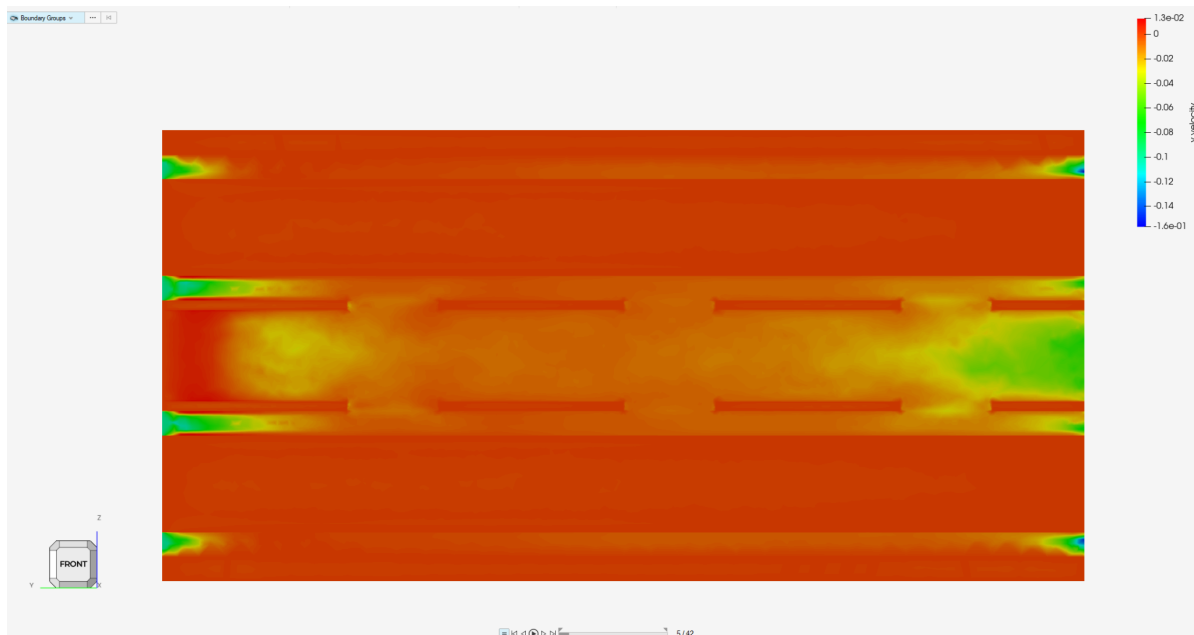


Figure 20: Lengthwise Slice of Y velocity at Timestep 5

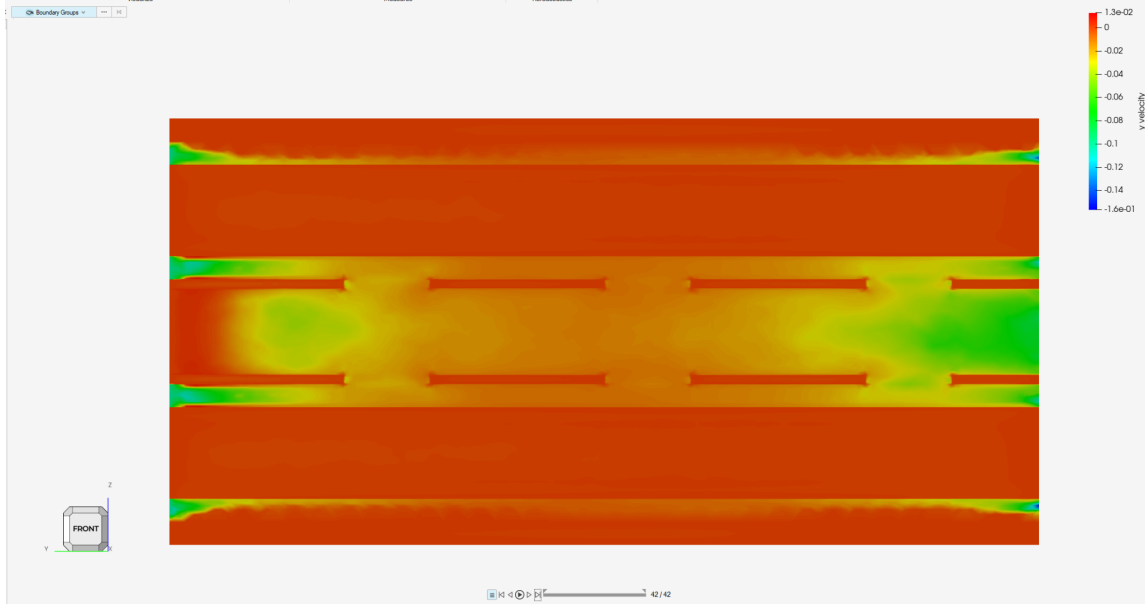


Figure 21: Lengthwise Slice of Y velocity at Timestep 42

The cross sectional y velocity plot shown in Figure 19 shows a relatively high velocity within the central channel. Turbulence due to the perforations are also observable. The velocity in the fluid between the strands is higher near the central channel and slower near the bounding box. Figures 20 and 21 show a clear evolution of the velocity in time. In these two figures, the inlets are on the left while the outlets are on the right. The results near the outlets should be disregarded due to not simulating backflow for computational efficiency.

5: Analysis and Closing Remarks (100% Rolleri)

This baseline steady state simulation was conducted to assess the accuracy of the model in its ability to produce physically sound results. Simplifications were made that had a significant effect on the results. These simplifications included the no slip wall on the inlet side of the fluid surrounding the strand and modelling incoming velocity as a constant. While the results of this

simulation do not accurately portray the temperatures present in CICs, the velocity seen within the design does show that the flow is highly turbulent and clearly moving between the central channel and the fluid domain between the strands. Therefore, to rectify these simplifications and model the system more accurately, a transient UDF simulation was run.

The thermal results of the transient UDF simulation were highly accurate given the geometric simplifications made due to computational constraints. The results clearly showed that the coolant removed the heat produced by the strands fast enough to maintain the entire system at 4 K. The monotone blue images depicting the temperature gradient within the CIC were expected. To further put the temperature results in perspective, real world tokamak CICs are roughly 400 meters in length [11]. Over that length, the helium does not increase in temperature by more than 1.5 K [12]. The model used here is 0.1 meters in length. Therefore, the temperature of the liquid helium from the inlet to the outlet was not expected to be greater than roughly 0.000375 K.

The velocity results of the transient UDF simulation were also highly accurate. However, backflow was not applied to the outlets of the simulation which did have an observable effect on the flow near the outlets. This simplification was made to decrease the simulation runtime. Nevertheless, Figures 19 - 21 show physically accurate results. Figure 19 clearly shows how the central channel is imparting velocity on the fluid between the central channel sleeve and the strands. The velocity of the fluid then quickly decreases between the strands as they act as a barrier to fluid entry. It is also important to note that the fluid here is highly turbulent leading to y velocity values significantly smaller than those assigned to the inlets, the x and z will each have some component of the overall velocity vector. Figures 20 and 21 show the evolution of velocity from the start to the end of the simulation. Propagation of the sinusoidal UDF is not easily

identifiable due to the fluid being modelled as incompressible. The major indicator of the presence of the UDF is the fact that the fluid velocity in the central channel is significantly higher than throughout the rest of the model.

When comparing the transient UDF results with the baseline steady state simulation results, it is obvious that CIC simulations must be set up to include flow in the strand region. In the baseline steady state simulation, applying a no slip BC to the inlet of the fluid around the strands clearly had a significant impact on heat removal within the system. This simulation was conducted to provide the group with insight into how the flow behavior of the central channel would interface with the fluid region in between the strands. The thermal and velocity plots, while insightful, do not accurately portray real world physics. However, as previously discussed, the transient UDF simulation does show results that are physical.

Going forward, the accuracy of the results from the UDF simulation could be improved. For example, the porosity could be reduced from 0.5 to 0.37. More strands could be added to increase the surface area in contact with the cryogenic helium coolant. Also, radiation could be incorporated into the simulation to mimic the heat deposited by the neutron flux produced within the reactor chamber. Neutron flux was neglected for this simulation because it does not contribute to heating with the CIC nearly as much as AC losses. However, for accuracy, this additional thermal load would need to be taken into account. The length of the CIC simulated could also be improved. Due to computational constraints, a length of the CIC only 100 mm long, or about 0.025% of the total actual length, was simulated. By simulating a larger section, a more significant thermal gradient could be observed directly rather than linearly extrapolating values as was done here. Lastly, the perforations, which for this analysis was simplified to regular holes within the sheath around the central channel, could be modelled as a spiral

perforation. The spiral perforation would likely give more accurate velocity results and improve turbulent mixing.

Appendix

A. 3D Model Detailed Dimensions

The length of all cylindrical bodies is 100 mm.

Body	Diameter (mm)
Central fluid	10
Central solid	12
Outer fluid	43.7
Magnetic strand	10.5

Table 4: Overall Model Dimensions

The bounding box had a height and width of 49 mm and a length of 100 mm. The windows on the central solid body spanned a 45 degree angle, were 10 mm in length, and spaced 5 mm apart. Two windows were placed on opposite sides of the central solid. Consecutive window sets were placed 45 degrees apart. The first window is 15 mm from the fluid inlet to ensure developed flow.

B. All Transient UDF Run Settings

Transient, SST $k - \omega$, 0.0156231 timestep, 1.25 sec. total runtime, 80 timesteps written out in intervals of 2 (+2 for first and last) - total: 42, heat transfer on, acuSolve (serial; no Intel MPI).

For UDF, applied values (in order):

- $V_0 = 0.5 \frac{m}{s}$
- $R = 0.005 \text{ m}$
- $\gamma = 157.6$
- $\nu = 0.0000000253 \frac{m^2}{s}$

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